MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

END OF SEMESTER EXAMINATION

ICT 1102 : INTRODUCTION TO PROGRAMMING

TIME ALLOWED: 3 HOURS **DATE : 14TH SEPTEMBER , 2017**

INSTRUCTIONS

i. This paper contains **TWO** pages. Please check.
ii. There are **Twelve Questions**in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

1. Define the following terms:
   a. Compiler. **[2 marks]**
   b. Interpreter. **[2 marks]**
   c. Syntax. **[2 marks]**
   d. Reserved word. **[2 marks]**
   e. Variable. **[2 marks]**
2. Study the pseudo-code for the program below and answer the questions that follow:

```
Program start
Use variables: value OF TYPE Integer
        letter OF TYPE Character
DISPLAY "Enter Letter"
ACCEPT letter
DISPLAY "Enter a positive number"
ACCEPT value
WHILE value <> 0 DO
        PRINT letter
        value:=value – 1
ENDWHILE
DISPLAY "Thank You"
Endprogram
```

   a. Explain how the pseudo-code for the program works. **[5 marks]**
   b. What will be the output from the program if a user enters **H**as a letter and **3**as a positive number? **[4 marks]**
3. Explain any **two** factors that can be considered when evaluating programming languages for use in software development. **[4 marks]**
4. What is the main difference between the while loop and the repeat until loop? **[4 marks]**
5. Write a pseudo code for a program that calculates and print the sum of numbers from 5 to 25. **[4 marks]**
6. Consider the following pseudo-code program segment that allows entry of a number in the range of 0 to 100.

```
Program start
Use variables: number OF TYPE Integer
REPEAT
        DISPLAY "Enter a number between 0 and 100"
        ACCEPT number
UNTIL number < 0 or number > 100
end program
```

   a. Draw a *flowchart* diagram to show this algorithm. **[3 marks]**
   b. Rewrite the pseudo-code for the program segment above using the *WHILE..DO* loop. **[4 marks]**

7. Write a pseudo code for a program that calculates and print the arithmetic mean of odd numbers in the range of 0 to 21. **[6 marks]**

8. Write a pseudo-code for a program that will calculate and print the result from a quadratic expression; $a^2 + b^2 - z^2 + q$; where the user has to provide a, b, z and q from the standard input device. **[4 marks]**

9. Write a pseudo code for a program that will print a series of numbers from 10 to 3 in reverse order. Use a While Loop. **[5 marks]**

10. Write a pseudo code for program that will calculate and print the cube of a number entered by a user until the user enters a number which is less than zero or more than 100. **[6 marks]**

11. Write a pseudo code for aprogram to determine whether a number entered through the keyboard is even or odd. **[5 marks]**

12. Explain the following logical operators:

    a. OR. **[3 marks]**

    b. AND. **[3 marks]**

    c. NOT. **[3 marks]**

END OF QUESTION PAPER